MILITARY SPECIFICATION

RESIN, EPOXY, LOW VISCOSITY

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of low viscosity epoxy resin.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipping and Storage
MIL-STD-1218	ACS Chemicals

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, Maryland 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1652-73 Epoxy Content of Epoxy Resins

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 Transportation

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)").

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The material shall be a low viscosity epoxy resin.

3.2 Chemical and physical properties. The chemical and physical properties shall be in accordance with TABLE I.

3.3 Workmanship. The material shall be uniform, free from contaminants, foreign material or any other defect that would prevent its use for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

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TABLE I. Chemical and physical properties.

Property	MIN	MAX
Weight per epoxide equivalent (g of resin containing one gram-equivalent -weight of epoxide) g	140	170
Water content, wt%		0.6
Viscosity, cps, 25°C	120	225
Specific gravity, 25/25°C	1.22	1.24
Refractive index at 25°C	1.4750	1.4850

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum

4.3 Sampling.

4.3.1 Lot. Unless otherwise specified in the contract (see 6.2), a lot shall consist of all material manufactured in one continuous production run or in one batch, under essentially identical conditions, from the same raw materials, and to be offered for acceptance at one time. Several batches, manufactured from the same raw materials, may be blended to form a uniform larger batch which shall then constitute a lot for inspection purposes.

4.3.2 Sampling. Sampling for quality conformance inspection shall be in accordance with inspection level I of MIL-STD-105. The sample unit shall be one unit package or container of material. Each sample shall consist of sufficient material to perform the quality conformance tests as specified in 4.4.

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4.4 Quality conformance inspection. Each sample obtained in accordance with 4.3.2 shall be subjected to the tests of 4.5. The acceptable quality level (AQL) shall be 2.5% defective. When specified in the contract or order (see 6.2), the contractor shall furnish test reports showing quantitative results for all quality conformance tests specified for each lot of material.

4.5 Tests. Unless otherwise specified herein all chemicals shall be ACS grade in accordance with MIL-STD-1218.

4.5.1 Weight per epoxide equivalent. The weight per epoxide equivalent shall be determined in accordance with ASTM D 1652-73.

4.5.2 Water content. The weight percent (wt%) water content shall be determined in accordance with the following:

4.5.2.1 Apparatus.

a. Beckman Aquameter, Beckman Instruments, Inc., Fullerton, California, or equivalent.

b. Lunge weighing pipette.

4.5.2.2 Procedure.

a. Place approximately 50 millilitres (ml) of absolute methanol into the reaction vessel of the Aquameter. Neutralize the methanol with Karl Fischer reagent.

b. Fill the Lunge pipette approximately one-half full with the sample. Close the stopcocks and weigh to the nearest 0.1 mg.

c. Pipette 50 to 75 drops (2 to 3 ml) of sample to the aquameter reaction vessel. As soon as the sample has dissolved in the neutralized methanol, titrate with standardized Karl Fischer reagent (one for which the grams of water corresponding to 1.0 ml of reagent has been determined).

NOTE: Using absolute methanol, clean and rinse electrodes after each sample has been run.

d. Reweigh the pipette, then calculate the weight of sample taken.

e. Calculate the wt% water content as follows:

$$\text{Water content, wt\%} = 100\,\frac{VE}{W}$$

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where: V = volume of Karl Fischer reagent used for titrating sample, ml
E = grams of water equivalent to 1.0 ml of Karl Fischer reagent used
W = weight of sample, g

4.5.3 Viscosity. The viscosity shall be determined in accordance with the following:

4.5.3.1 Apparatus.

a. Brookfield Syncho-lectric Viscometer, Model L.V.F., Brookfield Engineering Company, Inc., 240 Cushing Street, Stoughton, Massachusetts, or equivalent.

b. Water bath capable of maintaining a temperature of 25 ± 0.1°C.

c. A 600 ml beaker.

4.5.3.2 Procedure.

a. Place the sample in the beaker and temperature condition it in the water bath to 25 ± 0.1°C.

b. Attach a No. 1 spindle to the viscometer.

c. Lower the spindle into the sample so that the resin surface is at the machined groove in the spindle. Center the spindle in the beaker. Care must be exercised to prevent bubble entrapment on the spindle. Using the leveling device, level the viscometer.

d. Depress the clutch at the back of the viscometer, turn the instrument on; then release the clutch.

e. Allow the spindle to rotate at 30 revolutions per minute (rpm) for approximately one minute. Record the appropriate scale reading. Depress the clutch, and stop the motor. Obtain three readings and take the average.

f. Multiplying by the appropriate factor (as given in the tables and instructions furnished with the viscometer) convert the average scale reading for the spindle used into centipoise.

4.5.4 Specific gravity. The specific gravity, 25/25°C, shall be determined in accordance with the following:

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4.5.4.1 Apparatus.

a. Westphal specific gravity balance, or equivalent.

b. A 5 g displacement plummet. (It contains a thermometer and displaces exactly 5.000 g of distilled water at 25°C and is equipped with a thin platinum wire for attaching to the hook at the end of the balance beam.)

c. Balance weights. (The weights are 5, 0.5, 0.05, and 0.005 g. The balance beam is notched into units; when the weights are placed on the beam, they correspond to a specific gravity of 0.1, 0.01, 0.001, and 0.0001 for each unit of the beam.)

4.5.4.2 Procedure.

a. Mount the instrument; attach the plummet to the hook at the end of the balance beam, then adjust the screw at the base of the instrument until the beam rests at equilibrium.

b. Fill the sample cylinder with distilled water. Submerge the plummet (while still hanging from the balance beam) in the water making sure that it is completely submerged and that the twisted part of the platinum wire is also below the water level. Place the 5 g weight on the same hook with the plummet (that is, corresponding to notch No. 10 of the beam), then turn the adjustment screw to bring balance to perfect equilibrium again. The instrument is now ready for specific-gravity determinations.

c. Replace the water with sample having the same temperature as the water. Dry the plummet, then place it in the sample as before. Now balance the beam by placing weights on the beam until equilibrium is reached.

4.5.4.3 Calculation. Read the specific gravity by starting with the 5 g weight. Multiply the specific gravity unit number corresponding to this weight (0.1 for the 5 g weight) by the number of the notch on which it rests. Repeat this procedure with the rest of the weights and add the results to obtain the specific gravity.

EXAMPLE

Suppose that the following weights bring the balance to equilibrium:

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5 g weight on the hook (Notch No. 10)
0.5 g weight on notch No. 7
0.05 g weight on notch No. 6
0.005 g weight on notch No. 2

Specific gravity would be computed as follows:

0.1 (10) + 0.01 (7) + 0.001 (6) + 0.0001 (2)

Specific gravity = 1.0762 at 25/25°C

4.5.5 Refractive index. The refractive index at 25°C shall be determined in accordance with the following:

4.5.5.1 Apparatus. Abbe refractometer, or equivalent.

4.5.5.2 Procedure.

a. Maintain the instrument temperature at 25 ± 1°C for 10 to 15 minutes.

b. Clean the prisms with xylene.

c. Place one or two drops of sample on the lower prism. Close the prisms by means of the screw head.

d. Using the adjustment screw on the left side of the instrument, move the moving arm until the field of vision shows the boundary between light and shade just intersecting the crosshair.

e. Turn the adjustment screw on the right side of the instrument until the boundary between light and shade is as sharp as possible. Readjust the left hand screw if necessary.

f. Using the lens, read the refractive index directly from the sector. Repeat the adjustment three times. Report (to the fourth decimal place) the average of the values obtained.

4.6 Packaging inspection. The packaging, packing and marking shall be inspected to verify conformance with the requirements of section 5.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract or order (see 6.2), packaging and packing shall be level C.

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5.1.1 Level C. Unless otherwise specified in the contract or order (see 6.2), packaging and packing of epoxy resin shall be in accordance with standard commercial practice applicable to the type of material. The packaging and packing shall be of such construction and materials that the contents will be adequately protected against loss or contamination. Container size shall be as specified in the contract or order (see 6.2). Containers shall be capable of being resealed. Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation and shall be suitable for indoor storage at 32°C (90°F) maximum for a minimum of 2 years.

5.2 Marking. In addition to any special marking required by the contract or order (see 6.2), each container shall be marked in accordance with MIL-STD-129 and Code of Federal Regulations 49 CFR 171-179. Marking shall include but not be limited to, the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and location.

c. Material trade name.

d. Net weight and volume.

e. Lot number, batch number(s), and date of manufacture.

f. Storage conditions.

g. Contract or purchase order number.

6. NOTES

6.1 Intended use. Epoxy resin in accordance with this specification is intended for use as a curative agent in solid propellant formulations for the CKU-5/A rocket catapult assembly and other rocket motor applications.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Quantity required.

c. Place of delivery.

d. Inspection conditions when other than as specified (see 4.2).

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e. Lot size if other than as specified (see 4.3.1).

f. Packaging requirements if other than as specified (see 5.1).

g. Size of container required (see 5.1.1).

h. Any special marking required (see 5.2).

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID*
4.4	Quality conformance inspection data	-

*DID's (Data Item Description/DD Form 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DID's when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Suggested source of supply. A product that has met the requirements of this specification in past procurement actions is EPON 812, manufactured by the Miller Stephenson Chemical Company, 1350 Fullerton Avenue, Chicago, IL 60614. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

Custodian:
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Project Number:
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DOCUMENT IDENTIFIER AND TITLE

MIL-R-82664(OS), "Resin, Epoxy, Low Viscosity"

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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